

Nathan deBar

CS-499

Prof. Ogoh

### Milestone Two: Software Engineering Narrative

The artifact that I have selected to cover this enhancement is the Grazioso Salvare Animal Shelter CRUD application that I developed during CS-340 in August of 2025. This application was built to manage the records of animal shelters through a MongoDB database via a python script. It gives the users basic operations to interact with the database such as create, read, update, and delete. Additionally, the user can filter the data to help identify animals that are suitable candidates for rescue dog training programs.

I chose this artifact because it showcases an application that holds real world value and displays several software engineering principles surrounding database interactions, client and server communications, and data management. This alone is a strong foundation for enhancement as it already hosts functional code with ample room for improvement in its overall structure, reliability, and maintainability.

My focus for enhancement specifically revolves around the programs design as I intend to rework the structure to increase modularity. By separating the database operations, logic, and user interactions into clearly separated functions the code will have improved readability and maintainability. I have implemented this by adding a dedicated helper method like the connection handler and a reusable validation to reduce logic across the CRUD operations. Following this I intend to implement an input validation and error handling method to ensure that the application is capable of appropriately catching and handling invalid data or unexpected conditions clearly. I

managed to achieve this through validating the data types before database operations and by wrapping the interactions in try/except blocks. This increases the programs consistency and allows for safe fails. These changes bring the artifact in line with the course expectations and professional software engineering practices.

These enhancements support the course outcomes on multiple fronts. The initial restructuring shows the ability to design and evaluate code against appropriate software engineering practices. By adding validation and error handling I am also displaying proper use of techniques and tools to ensure that the system remains reliable and maintainable. Additionally, these changes show that I have a security mindset to ensure programs I am working on follow appropriate security development practices. The validations prevent the wrong type of data from being processed and reduces the risk of unintended behaviors from occurring, which falls in line with the secure coding practices that were listed in the course outcomes.

I have gained a far stronger understanding of appropriate structure in relation to its impact on maintainability and scalability. When refactoring the code, I had to consider how each of the key components interact with one another and what the most logical and sensible method to dividing the responsibilities should be. A major challenge I ran into during this refactor was not breaking the overall logic when separating these methods. I had to ensure that the database operations continued to function which led to testing the code to ensure that as each part was moved that it still held its integrity. Overall, the enhancements that I have made have shifted the project from a prototype into a more structured and maintainable system while reflecting industry standards for software engineering.